

Many powerful numbers between consecutive powers

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Abstract

Let $\kappa \geq 2$ and let $h_\kappa(n)$ denote the number of κ -full integers in the open interval $(n^\kappa, (n+1)^\kappa)$. De Koninck and Luca, and De Koninck–Luca–Shparlinski, proved that $h_\kappa(n) \geq (\log n)^{1/3+o(1)}$ for infinitely many n . We show by a refinement of their method — replacing a self-referential approximation condition by a fixed-tolerance application of Dirichlet’s theorem, and choosing the auxiliary squarefree numbers as the divisors of a primorial — that for every fixed $\kappa \geq 2$,

$$h_\kappa(n) \gg_\kappa \frac{\log n}{\log \log n \log \log \log n}$$

for infinitely many n . In particular, in Erdős’s question whether $h_2(n) \asymp (\log n)^{c \pm o(1)}$, any admissible exponent satisfies $c \geq 1$.

1 Introduction

An integer $m \geq 1$ is κ -full (for $\kappa = 2$, *powerful* or *squarefull*) if $p^\kappa \mid m$ for every prime $p \mid m$. Since the number of powerful integers up to x is $\sim \frac{\zeta(3/2)}{\zeta(3)} \sqrt{x}$ [1], the interval $(n^2, (n+1)^2)$ contains on average about $\frac{\zeta(3/2)}{\zeta(3)} - 1 \approx 1.17$ powerful numbers besides none being a perfect square. Erdős asked (see [2, Problem #942]) for the maximal order of

$$h_\kappa(n) := \#\{m \in (n^\kappa, (n+1)^\kappa) : m \text{ is } \kappa\text{-full}\},$$

and in particular whether there is a constant $c > 0$ with $h_2(n) < (\log n)^{c+o(1)}$ for all n and $h_2(n) > (\log n)^{c-o(1)}$ infinitely often.

De Koninck and Luca [3] proved $h_2(n) > \frac{9}{20}(\log n / \log \log n)^{1/3}$ for infinitely many n ; De Koninck, Luca and Shparlinski [4] extended this to every fixed $\kappa \geq 2$ (indeed uniformly in $\kappa = \kappa(N)$ growing slowly); we are not aware of a previous improvement of these lower bounds.

A related line of work studies the *distribution* of h_κ rather than its maximal order. Shiu [5] proved that for each fixed ℓ the set of n for which $(n^2, (n+1)^2)$ contains exactly ℓ non-square powerful integers has a positive asymptotic density, and Xiong and Zaharescu [6] extended this to κ -full integers between successive κ th powers (recently Narumi and Tachiya [7] treated adjacent such intervals). These are fixed- ℓ density statements; they do not bound $h_\kappa(n)$ for ℓ growing with n , which is the regime studied here. We prove:

Theorem 1.1. *For every fixed $\kappa \geq 2$ there is $c_\kappa > 0$ such that*

$$h_\kappa(n) \geq c_\kappa \frac{\log n}{\log \log n \log \log \log n}$$

for infinitely many positive integers n .

Thus any exponent c as in Erdős's question satisfies $c \geq 1$. The proof refines the method of [3, 4] in two ways. First, those papers impose on the common denominator q a self-referential approximation condition ($|\alpha_j - r_j/q| \leq q^{-1-1/2\ell}$ for all j , with q minimal), whose analysis requires Roth's theorem and yields $\log n \asymp \ell^3 \log \ell$ for ℓ constructed κ -full numbers; but the placement of the constructed numbers in the target interval only requires the *fixed* tolerance $\|q\alpha_j\| \leq c/(Dd_j^{1/\kappa})$, which Dirichlet's theorem supplies directly with $\log q \ll \ell \log(D^2)$, and Roth's theorem is not needed. Second, the auxiliary squarefree numbers d_j need not be the first 2ℓ squarefree integers with $D = \prod_j d_j$: it suffices that each d_j divides D , so choosing D to be a primorial and the d_j its squarefree divisors reduces $\log D$ from $\Theta(\ell \log \ell)$ to $O(\log \ell \log \log \ell)$, and the available upper bound for $\log n$ to $O_\kappa(\ell \log \ell \log \log \ell)$.

2 Proof of Theorem 1.1

Fix $\kappa \geq 2$ and a large integer ℓ . Let h be minimal with $2^h - 1 \geq 2\ell$, so $h \asymp \log \ell$, and let

$$D = p_1 p_2 \cdots p_h$$

be the product of the first h primes; by the prime number theorem $\log D \asymp h \log h \asymp \log \ell \log \log \ell$. Let $d_1 < \cdots < d_{2\ell}$ be any 2ℓ of the $2^h - 1$ squarefree divisors $d > 1$ of D , and set $\alpha_j = d_j^{-1/\kappa}$, an irrational number, for $j = 1, \dots, 2\ell$.

Choice of q

By the box principle (Dirichlet's theorem on simultaneous approximation, with the per-coordinate tolerances $\delta_j := \frac{1}{2^{\kappa+1}\kappa D d_j^{1/\kappa}}$): subdividing $[0, 1)^{2\ell}$ into $\prod_j \lceil \delta_j^{-1} \rceil$ boxes and considering the points $(\{t\alpha_1\}, \dots, \{t\alpha_{2\ell}\})$ for $0 \leq t \leq \prod_j \lceil \delta_j^{-1} \rceil$, there exist $t_1 > t_2$ in the same box; $q := t_1 - t_2$ satisfies

$$1 \leq q \leq \prod_{j=1}^{2\ell} \lceil \delta_j^{-1} \rceil \leq (4\kappa 2^\kappa D^{1+1/\kappa})^{2\ell}, \quad \|q\alpha_j\| \leq \delta_j \quad (j = 1, \dots, 2\ell). \quad (1)$$

In particular $\log q \ll_\kappa \ell \log D \ll \ell \log \ell \log \log \ell$.

Let r_j be the nearest integer to $q\alpha_j$ and $\varepsilon_j = q\alpha_j - r_j$, so $|\varepsilon_j| \leq \delta_j$ and $\varepsilon_j \neq 0$ (each $\alpha_j = d_j^{-1/\kappa}$ is irrational, as $d_j > 1$ is squarefree). Note $r_j \geq 1$: if $r_j = 0$ then $\|q\alpha_j\| = q\alpha_j \geq \alpha_j = d_j^{-1/\kappa} > \delta_j$, a contradiction.

The constructed numbers

Set $n = Dq$ and

$$m_j = d_j D^\kappa r_j^\kappa \quad (j = 1, \dots, 2\ell).$$

Each m_j is κ -full: every prime of d_j divides D , hence appears in m_j with exponent at least $\kappa + 1$, while primes of D or of r_j appear with exponent at least κ . The m_j are pairwise distinct: comparing p -adic valuations modulo κ in $d_i r_i^\kappa = d_j r_j^\kappa$ forces $v_p(d_i) = v_p(d_j)$ for all p , i.e. $d_i = d_j$.

Placement

Since $d_j \alpha_j^\kappa = 1$,

$$d_j r_j^\kappa = d_j (q \alpha_j - \varepsilon_j)^\kappa = q^\kappa - \kappa q^{\kappa-1} d_j^{1/\kappa} \varepsilon_j + E_j, \quad |E_j| \leq 2^\kappa q^{\kappa-2} d_j^{2/\kappa} \varepsilon_j^2 \kappa^2,$$

say, for q large. Hence

$$m_j - n^\kappa = D^\kappa (d_j r_j^\kappa - q^\kappa) = -\kappa D^\kappa q^{\kappa-1} d_j^{1/\kappa} \varepsilon_j + D^\kappa E_j.$$

Write $M_j := \kappa D^\kappa q^{\kappa-1} d_j^{1/\kappa} |\varepsilon_j|$ for the absolute value of the main term. With $|\varepsilon_j| \leq \delta_j = (2^{\kappa+1} \kappa D d_j^{1/\kappa})^{-1}$ we get $M_j \leq 2^{-(\kappa+1)} D^{\kappa-1} q^{\kappa-1} = 2^{-(\kappa+1)} n^{\kappa-1}$, while the error is dominated termwise:

$$\frac{D^\kappa |E_j|}{M_j} \leq \frac{2^\kappa \kappa^2 q^{\kappa-2} d_j^{2/\kappa} \varepsilon_j^2}{\kappa q^{\kappa-1} d_j^{1/\kappa} |\varepsilon_j|} = \frac{2^\kappa \kappa d_j^{1/\kappa} |\varepsilon_j|}{q} \leq \frac{1}{2Dq} \leq \frac{1}{2}.$$

Hence $\frac{1}{2} M_j \leq |m_j - n^\kappa| \leq \frac{3}{2} M_j$; since $\varepsilon_j \neq 0$ this gives

$$0 < |m_j - n^\kappa| \leq \frac{3}{2} \cdot 2^{-(\kappa+1)} n^{\kappa-1} < n^{\kappa-1} \leq \min((n+1)^\kappa - n^\kappa, n^\kappa - (n-1)^\kappa),$$

and the sign of $m_j - n^\kappa$ equals that of the main term, i.e. is opposite to that of ε_j . Therefore each m_j lies in $(n^\kappa, (n+1)^\kappa)$ if $\varepsilon_j < 0$, and in $((n-1)^\kappa, n^\kappa)$ if $\varepsilon_j > 0$. By the pigeonhole principle, at least ℓ of the 2ℓ numbers m_j lie in a single one of these two intervals; setting $N = n$ in the first case and $N = n - 1$ in the second,

$$h_\kappa(N) \geq \ell.$$

Conclusion

Let $X = \log N$ and $f(x) = x/(\log x \log \log x)$; one checks f is increasing for $x \geq 16$. We have shown $X \leq Y_\ell := C_\kappa \ell \log \ell \log \log \ell$ with $C_\kappa \geq 1$, and $X \rightarrow \infty$ as $\ell \rightarrow \infty$ (note $N \geq D - 1$). Since $Y_\ell \geq \ell$, we have $\log Y_\ell \geq \log \ell$ and $\log \log Y_\ell \geq \log \log \ell$, so monotonicity gives, for ℓ large,

$$f(X) \leq f(Y_\ell) = \frac{C_\kappa \ell \log \ell \log \log \ell}{\log Y_\ell \log \log Y_\ell} \leq C_\kappa \ell, \quad \text{i.e.} \quad \ell \gg_\kappa \frac{\log N}{\log \log N \log \log \log N}.$$

Letting $\ell \rightarrow \infty$ produces infinitely many such N . $\square$

Remark 2.1. The construction is completely explicit and was verified computationally in small cases. For $\kappa = 2$, $h = 2$, $D = 6$: a small search finds $q = 485$, giving $N = 2909$, and two of the constructed powerful numbers lie in $(2909^2, 2910^2)$. For $h = 3$ ($D = 30$, all seven divisors $2, 3, 5, 6, 10, 15, 30$): a lattice-reduction search finds q with about 26 decimal digits for which five of the seven constructed numbers lie in $(n^2, (n+1)^2)$ (and two in the lower interval). Theorem 1.1 has been formalized and machine-checked in Lean 4/Mathlib, for every fixed $\kappa \geq 2$, with no `sorry` and only the standard axioms. The development certifies the simultaneous box principle, the κ -fullness and distinctness of the constructed numbers $d_j D^\kappa r_j^\kappa$ (the smallest case $\kappa = 2$, $D = 6$ places $8467200 = 3 \cdot 6^2 \cdot 280^2$ and $8468064 = 6 \cdot 6^2 \cdot 198^2$ in $(2909^2, 2910^2)$), the window placement, the count $2^{\omega(D)} - 1$ of squarefree divisors of the primorial, and the size bound $\log \prod_{i < h} p_i \ll h \log h$ — which uses the Chebyshev lower bound $\pi(x) \gg x/\log x$ to bound the h th prime by $O(h \log h)$ — together with the pigeonhole count and the final inversion, yielding $c_\kappa \log n / (\log \log n \log \log \log n) \leq h_\kappa(n)$ for infinitely many n . The certificate is available at <https://github.com/scottdhughes/erdos942>.

Remark 2.2. The bottleneck is the box principle: tolerance $e^{-\Theta(\log \ell \log \log \ell)}$ in dimension 2ℓ costs $\log q \asymp \ell \log \ell \log \log \ell$, so Theorem 1.1 may be close to the natural limit of this Dirichlet/primorial construction; removing the remaining $\log \log N \log \log \log N$ factor appears to require either simultaneous approximation of quality beyond Dirichlet's for the specific algebraic vector $(d_j^{-1/\kappa})_j$, or a different construction.

Use of AI

Claude Opus 4.8, Codex (GPT-5.5), and Harmonic's Aristotle were used in the preparation of this work. All Lean-formalized results are verified by the Lean kernel and depend only on the standard axioms, and the author has checked the mathematics and the cited literature and takes responsibility for all claims.

References

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